

CSE 207

B.Tech. IVth SEMESTER EXAMINATION, 2024-25

BACHELOR OF TECHNOLOGY

(CSE)

Operating Systems

5919

Time : Three Hours]

[Maximum Marks : 75

Note: There are three sections (A, B and C) and Candidate has to attempt question from all sections. Marks are Indicated against each section .

Section-A

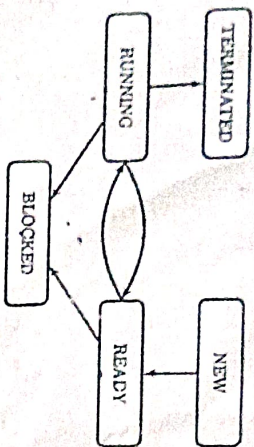
1. Attempt all parts of the following : $5 \times 3 = 15$

(a) A computer has 6 tape drives and n processes are competing for them. Each process may need two drives to complete its execution. What is the maximum and minimum value of n for which the system will be always deadlock free.

~~(b)~~ What do you mean by convoy effect?

~~(c)~~ Differentiate the resource portioning approach and pool based approach for resource allocation.

(d) Consider the process state transition diagram in the figure given below :



Which type of CPU scheduling algorithm is being used here ?

(e) What do you mean by mutual exclusion ?

Section-B

Note: Attempt All Questions :

4×5=20

2/ (a) Consider the following segment table :

Segment	Base	Length
0	219	600
1	2300	14
2	90	100
3	1327	580
4	1952	96

What are the physical addresses for the following logical addresses?

- | | |
|------------|-----------|
| (i) 0430 | (iv) 3400 |
| (ii) 110 | (v) 4112 |
| (iii) 2500 | |

OR

(b) Describe the paging scheme used in operating system using diagram. Consider a logical-address space of eight pages of 1024 words each, mapped onto a physical memory of 32 frames.

(i) How many bits are in the logical address?

(ii) How many bits are in the physical address?

(a) Show that TLB reduces the EAT (Effective Access Time) in a paging based system with suitable diagram. Consider a paging system with the page table stored in memory .

(i) If a memory takes 200 Nano seconds, how long does a paged memory reference take ?

(ii) If we add TLBs and 75% of all page table references are found in the TLBs, what is the effective memory reference time? (Assume that finding a page-table entry in the TLBs takes zero time-if the entry is there .)

OR

- (b) Differentiate external fragmentation and internal fragmentation with suitable diagrams. Consider a paging system where page size is 2KB. There are four processes as: P_0 , P_1 , P_2 , and P_3 with size of 150KB, 351KB, 225KB and 601KB.

Calculate the total no. of pages required for P_0 , P_1 , P_2 , and P_3 with the total internal fragmentation.

4. ✓ (a)

Let a memory have four free blocks of sizes 4K, 8K, 20K and 2K. These blocks are allocated following the best-fit strategy. The allocation requests are stored in a queue as shown below :

Request No	Request Sizes	Usage Time
J_1	2K	4
J_2	14K	10
J_3	3K	2
J_4	6K	8
J_5	6K	4
J_6	10K	1
J_7	7K	8
J_8	20K	6

Compute the time at which the request for J_7 will be completed.

OR

- (b) The address sequence generated by tracing a particular program executing in a pure demand paging system with 100 records per page, with 1 free main memory frame is recorded as follows :
- 0100, 0200, 0430, 0499, 0510, 0530, 0560, 0120, 0220, 0240, 0260, 0320, 0370 .

5. ✓ (a) Compare Monolithic and Micro Kernel approach of Kernel design with suitable diagram .

OR

- (b) Compare process against thread with suitable diagram.

Section-C

Note: Attempt Any Two Questions :

2×20=40

6. (a) Describe Peterson's solution with suitable steps implemented in C-language .

(b) What are the various performance indicators for CPU scheduling algorithms? Describe in detail. Consider 'n' processes sharing the CPU in a round-robin fashion. Assuming that each process switch takes 's' seconds, what must be the quantum size 'q' such that the overhead resulting from process switching is minimized but at the same time each process is guaranteed to get its turn at the CPU at least every 't' seconds?

7. (a) What are the various conditions for deadlock? Consider the Allocation Matrix and Max Matrix of resources X, Y and Z for the processes P_0 , P_1 and P_2 as :

	Allocation			Max		
	X	Y	Z	X	Y	Z
P_0	0	0	1	8	4	3
P_1	3	2	0	6	2	0
P_2	2	1	1	3	3	3

Show that currently this system is free from deadlock if there are 3 units of type X, 2 units of type Y and 2 units of type Z resources are still available.

Now consider the followings :

REQ 1: P_0 requests 0 units of X, 0 units of Y and 2 units of Z.

REQ 2: P_1 requests 2 units of X, 0 units of Y and 0 units of Z.

Which request, REQ1 or/and REQ2 will be granted?

(b) Describe page fault handling in demand paging with the help of suitable diagram.

A certain computer provides its users with a virtual memory space of 2^{32} bytes. The computer has 2^{18} bytes of physical memory. The virtual memory is implemented by paging, and page size is 4096 bytes. A user process generates the virtual address 11123456. Explain how the system establishes the corresponding physical location.

8. (a) Consider the reference string given as :
7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1, 7,
0, 1 . find out page fault rate using FIFO and
optimal page replacement algorithm. Use 3
frames to compute your result .
- (b) Explain the followings :
(i) Valid and invalid bit for memory
protection.
(ii) Shared pages
9. (a) Consider a disk queue with requests for I/O
to blocks on cylinders as :
98, 183, 37, 122, 14, 124, 65, 67 .
Disk head is initially at cylinder 53.
Count total head movements using FcFs disk
scheduling and SSTF disk scheduling
approaches.
- (b) Discuss Dining-Philosopher problem with
suitable diagram and propose its solution.

••••